

2 Subject content and assessment information

The GCSE in Design and Technology enables students to understand and apply iterative design processes through which they explore, create and evaluate a range of outcomes. The qualification enables students to use creativity and imagination to design and make prototypes (together with evidence of modelling to develop and prove product concept and function) that solve real and relevant problems, considering their own and others' needs, wants and values. It gives students opportunities to apply knowledge from other disciplines, including mathematics, science, art and design, computing and the humanities.

Students will acquire subject knowledge in design and technology that builds on Key Stage 3, incorporating knowledge and understanding of different materials and manufacturing processes in order to design and make, with confidence, prototypes in response to issues, needs, problems and opportunities. Students learn how to take design risks, helping them to become resourceful, innovative and enterprising citizens. They should develop an awareness of practices from the creative, engineering and manufacturing industries. Through the critique of the outcomes of design and technology activity, both historic and present day, students should develop an understanding of its impact on daily life and the wider world and understand that high-quality design and technology is important to the creativity, culture, sustainability, wealth and wellbeing of the nation and the global community.

In the context of this document, the term 'prototype' refers to a functioning design outcome. A final prototype could be a highly-finished product, made as proof of concept before manufacture, or working scale models of a system where a full-size product would be impractical.

Qualification aims and objectives

The study of design and technology seeks to prepare students to participate confidently and successfully in an increasingly technological world. It helps students to be aware of, and learn from, wider influences on design and technology, including historical, social/cultural, environmental and economic factors.

The aims and objectives of this qualification are to enable students to:

- demonstrate their understanding that all design and technological activity takes place in contexts that influence the outcomes of design practice
- develop realistic design proposals as a result of the exploration of design opportunities and users' needs, wants and values
- use imagination, experimentation and combine ideas when designing
- develop the skills to critique and refine their own ideas while designing and making
- communicate their design ideas and decisions using different media and techniques, as appropriate for different audiences at key points in their designing
- develop decision-making skills, including the planning and organisation of time and resources when managing their own project work
- develop a broad knowledge of materials, components and technologies and practical skills to develop high-quality, imaginative and functional prototypes
- be ambitious and open to explore and take design risks in order to stretch the development of design proposals, avoiding clichéd or stereotypical responses
- consider the costs, commercial viability and marketing of products

- demonstrate safe working practices in design and technology
- use key design and technology terminology, including those related to: designing, innovation and communication; materials and technologies; making, manufacture and production; critiquing, values and ethics.

Mathematics

Maths skills are fundamental to design and technology. These skills will be embedded in the examination for this qualification. Please see *Appendix 1: Mathematical skills* for full details of these maths skills and examples of their application.

Please see *Appendix 2: Calculators* for information on the use of calculators in the assessments of this qualification.

Scientific skills, knowledge and understanding

Science skills, knowledge and understanding underpin the theory and practice of design and technology. These skills will be embedded in the examination for this qualification. Please see *Appendix 3: Science skills, knowledge and understanding* for full details of these skills and examples of their application.

Links to other relevant subjects

In addition to maths and science, the GCSE in Design and Technology embeds knowledge, techniques and practices from a wide range of subjects, including art and design, business, computer science and geography (see *Appendix 4*). As part of the teaching of the qualification, teachers should use *Appendix 4* with the identified links to the subject content, to ensure that students gain an awareness and appreciation of the ways in which these subject areas relate to and inform decisions in design.

CORE

Component 1

Overview

The content is divided into two sections: core content and material categories. The core content provides students with a breadth of study and the material categories provides students with a depth of study.

All students must study the Core content plus at least **one** material category.

All topics within the core content and the chosen material category must be covered but can be delivered in any order or in an integrated approach.